

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0194

Los Angeles 1617 138 kV
September 30, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0194. The Interconnection Customer proposes a 181.8 MW solar photovoltaic facility interconnecting at the Los Angeles 1617 138 kV substation in the Virginia area. The request was received on January 05, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 128 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$198,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

VILLA MAR 1657 – LOS ANGELES 1617 SUBSTATION 138.0 kV: 100.8% loading (227 MVA) following the outage of the MICKENS SUBSTATION – WORKS 53 345.0 kV line; DFAX = 5.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VILLA MAR 1657 – LOS ANGELES 1617 138.0 kV	Reconductor line	\$1,170,000	17% (\$198,000)
	Total allocated to project		\$198,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.